

CHE 201 Spring 2020 HW 1

Roxanna Mendoza

TOTAL POINTS

100 / 100

QUESTION 1

1 Question 1 10 / 10

✓ - **0 pts** Correct

QUESTION 2

2 Question 2 20 / 20

✓ - **0 pts** Correct

- **3 pts** Wrong occurrences within the dishwasher system.

- **3 pts** Wrong occurrences within the shower system.

- **4 pts** Description could be more detailed.

- **3 pts** Missing an identification of the location of a system boundary.

- **3 pts** Wrong occurrences within the water heater system.

- **9 pts** Missing identifications of the locations of the system boundaries.

QUESTION 3

3 Question 3 10 / 10

- **3 pts** Answers of b,d,h are wrong. Formulas are right. Just recalculate.

✓ - **0 pts** Corret

- **1 pts** Can not read a)

- **1 pts** Answer of e) is wrong. Formulas are right. Just recalculate.

- **1 pts** Answer of h) is wrong.

- **0.5 pts** Answer of d) need to be more precise.

QUESTION 4

4 Question 4 10 / 10

✓ - **0 pts** Correct

- **5 pts** Redo calculation of circled parts.

- **5 pts** Did not answer 4 b)

- **10 pts** No answer

QUESTION 5

5 Question 5 10 / 10

✓ - **0 pts** Correct

- **5 pts** Unit of B is wrong.

- **0 pts** You lost a "f" on A

- **5 pts** Unit of B is wrong. $[B]=[P]^*[V]$

- **2.5 pts** So what's the unit of B?

- **10 pts** No answer submitted.

QUESTION 6

6 Question 6 10 / 10

✓ - **0 pts** Correct

- **5 pts** Wrong judgement on vacuum and gage pressure.

- **5 pts** Wrong reading of Bourdon gage.

- **10 pts** No answer submitted.

QUESTION 7

7 Question 7 10 / 10

- **2 pts** Wrong units of final results.

✓ - **0 pts** correct

- **3 pts** Wrong judgement of pressure increase or decrease.

- **3 pts** Did not answer the second question, increase or decrease of pressure.

- **7 pts** Did not answer the pressure difference between a and b.

- **10 pts** No answer.

QUESTION 8

8 Question 8 20 / 20

✓ - **0 pts** Well explained.

- **4 pts** Investigation on the physical basis of Korotkoff sound could be more detailed and specific.

- **4 pts** Investigation on the role of systolic and diastolic pressure on medical practice could be more detailed and specific.

- **2 pts** Investigation on the physical basis of Korotkoff sound could be more detailed and specific.

- **2 pts** Investigation on the role of systolic and diastolic pressure on medical practice could be more detailed and specific.

- **2 pts** No references listed.

- **20 pts** No answer.

- **8 pts** No answer of the significance of these pressures on medical practice.

- **10 pts** The handwriting is hard to read. Also, the investigation could be more detailed and specific.

Homework #1

Roxanna Mendoza

#1.

1.1 I. Boundary

1.2 G. Closed System

1.3 B. Control Volume

1.4 E. Extensive property

1.5 H. Intensive property

1.6 D. Process

1.7 J. Property

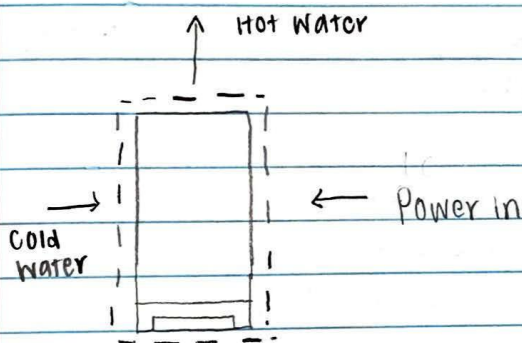
1.8 A. State

1.9 F. Surroundings

1.10 C. System

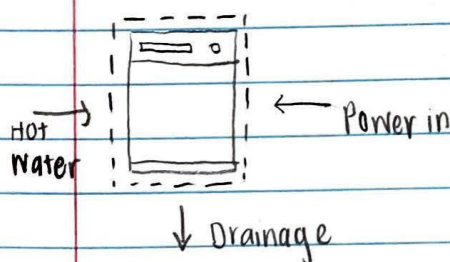
#2.

Hot Water Heater

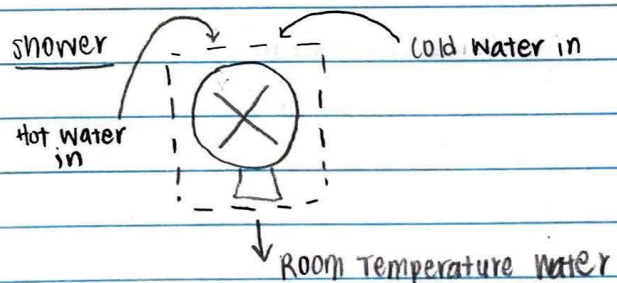


• The Hot water heater's boundary lies on its surface. Cold water flows into the system crossing the boundary and hot water then flows out with the assistance of the electric current that heats the water and passes through the system as well.

Dishwasher



• For a dishwasher the boundary lies on its surface. It is powered by electricity. Hot water goes into the system.



• For the shower the boundary lies on its surface. Both cold water and hot water go in and then the hot water exchanges heat with the cold water to meet an equilibrium temperature regulated by user.

1 Question 1 10 / 10

✓ - 0 pts Correct

Homework #1

Roxanna Mendoza

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1.2 G. Closed System

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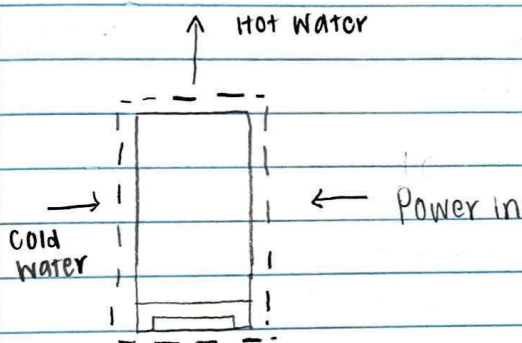
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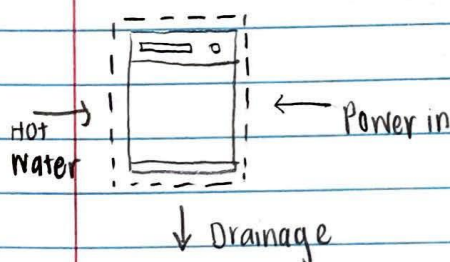
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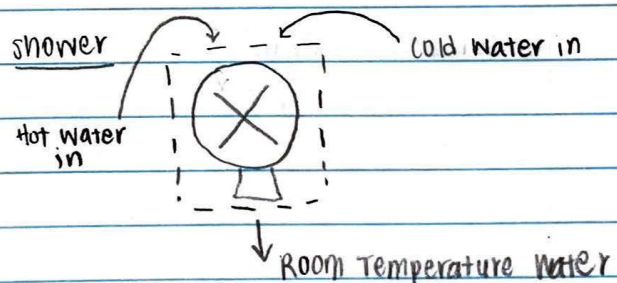


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2 Question 2 20 / 20

✓ - 0 pts Correct

- 3 pts Wrong occurrences within the dishwasher system.
- 3 pts Wrong occurrences within the shower system.
- 4 pts Description could be more detailed.
- 3 pts Missing an identification of the location of a system boundary.
- 3 pts Wrong occurrences within the water heater system.
- 9 pts Missing identifications of the locations of the system boundaries.

#3.

A. $122 \text{ in}^3 \Rightarrow \text{L}$

$$1 \text{ inch} = 0.0254 \text{ m} \quad \& \quad 1000 \text{ L} = 1 \text{ m}^3$$

$$\frac{122 \text{ in}^3}{1 \text{ inch}} \times \frac{(0.0254 \text{ m})^3}{1 \text{ in}^3} \times \frac{1000 \text{ L}}{1 \text{ m}^3} = 1.999 \text{ L}$$

B. $778.17 \text{ ft} \cdot \text{lb}_f \rightarrow \text{kJ}$

$$1 \text{ Newton} = 0.225 \text{ lb}_f \quad \& \quad 1 \text{ ft} = 12 \text{ inches} \quad \& \quad 1 \text{ in} = 0.0254 \text{ m}$$

$$\frac{778.17 \text{ ft} \cdot \text{lb}_f}{1 \text{ Newton}} \times \frac{1 \text{ Newton}}{0.225 \text{ lb}_f} \times \frac{12 \text{ inches}}{1 \text{ ft}} \times \frac{0.0254 \text{ m}}{1 \text{ in}} = 1054.10 \text{ J} = 1.054 \text{ kJ}$$

C. $100 \text{ hp} \rightarrow \text{kW}$

$$1 \text{ hp} = 550 \text{ ft} \cdot \text{lb}_f/\text{s}$$

$$\frac{100 \text{ hp}}{1 \text{ hp}} \times 550 \text{ ft} \cdot \text{lb}_f/\text{s} = 55,000 \text{ ft} \cdot \text{lb}_f/\text{s}$$

$$\frac{55,000 \text{ ft} \cdot \text{lb}_f/\text{s}}{1 \text{ Newton}} \times \frac{1 \text{ Newton}}{0.225 \text{ lb}_f} \times \frac{12 \text{ inches}}{1 \text{ ft}} \times \frac{0.0254 \text{ m}}{1 \text{ in}} = 74500.66 \text{ J} = 74.51 \text{ kJ}$$

D. $1000 \text{ lb/h} \rightarrow \text{kg/s}$

$$1 \text{ lb} = 0.454 \text{ kg} \quad \& \quad 60 \text{ s} = 1 \text{ min} \quad \& \quad 60 \text{ min} = 1 \text{ hr}$$

$$\frac{1000 \text{ lb}}{\text{h}} \times \frac{0.454 \text{ kg}}{1 \text{ lb}} \times \frac{1 \text{ hr}}{60 \text{ min}} \times \frac{1 \text{ min}}{60 \text{ sec}} = 0.126 \text{ kg/s}$$

E. $29.392 \text{ lb}_f/\text{in}^2 \rightarrow \text{bar}$

$$1 \text{ N} = 0.225 \text{ lb}_f \quad \& \quad 1 \text{ in} = 0.0254 \text{ m} \quad \& \quad 1 \text{ pascal} = 1 \times 10^{-5} \text{ bar}$$

$$\frac{29.392 \text{ lb}_f}{\text{in}^2} \times \frac{1 \text{ N}}{0.225 \text{ lb}_f} \times \left(\frac{1 \text{ in}}{0.0254 \text{ m}} \right)^2 = 202478.63 \frac{\text{N}}{\text{m}^2}$$

$$\frac{202478.63}{1 \text{ pascal}} \times 1 \times 10^{-5} \text{ bar} = 2.025 \text{ bar}$$

F. $2500 \text{ ft}^3/\text{min}$ to m^3/s

1 inch = 0.0254 m $\therefore$ 12 in = 1 ft $\therefore$ 60 s = 1 min

$$\frac{2500 \text{ ft}^3}{\text{min}} \left| \frac{(12 \text{ in})^3}{(1 \text{ ft})^3} \right| \left| \frac{(0.0254 \text{ m})^3}{(1 \text{ in})^3} \right| \left| \frac{1 \text{ min}}{60 \text{ s}} \right| = \frac{1.19 \text{ m}^3}{\text{s}}$$

G. 75 mile/h to km/h

1 mile = 5280 ft $\therefore$ 12 in = 1 ft $\therefore$ 1 in = 0.0254 m $\therefore$ 1000 m = 1 km

$$\frac{75 \text{ mile}}{\text{h}} \left| \frac{5280 \text{ ft}}{1 \text{ mile}} \right| \left| \frac{12 \text{ in}}{1 \text{ ft}} \right| \left| \frac{0.0254 \text{ m}}{1 \text{ in}} \right| \left| \frac{1 \text{ km}}{1000 \text{ m}} \right| = \frac{120.7 \text{ km}}{\text{h}}$$

H. 1 ton (= 2000 lbf) to N

1 N = 0.225 lbf

$$\frac{2000 \text{ lbf}}{1} \left| \frac{1 \text{ N}}{0.225 \text{ lbf}} \right| = 8888.89 \text{ N}$$

#4.

A. Octane C_8H_{18}

$12 \cdot 8 + 1 \cdot 18 = 114.23 \text{ g/mol}$ $\leftarrow$ molar mass of octane

0.3 kmol $\leftarrow$ convert to moles by multiplying by 1000

300 mol of octane

$$\frac{300 \text{ mol}}{1} \left| \frac{114.23 \text{ g}}{\text{mol}} \right| = 34269 \text{ g} \leftarrow \text{to kilograms divide by 1000}$$

$$= 34.269 \text{ kg}$$

kg to N $\leftarrow$ multiply by 9.81 m/s^2

$W = mg$ $\leftarrow$ we know force is equal to mass times gravitational force

$$W = (34.269 \text{ kg})(9.81 \text{ m/s}^2)$$

$$W = 336.19 \text{ N}$$

3 Question 3 10 / 10

- **3 pts** Answers of b,d,h are wrong. Formulas are right. Just recalculate.

✓ - **0 pts** Corret

- **1 pts** Can not read a)

- **1 pts** Answer of e) is wrong. Formulas are right. Just recalculate.

- **1 pts** Answer of h) is wrong.

- **0.5 pts** Answer of d) need to be more precise.

F. $2500 \text{ ft}^3/\text{min}$ to m^3/s

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$$\frac{2500 \text{ ft}^3}{\text{min}} \left| \frac{(12 \text{ in})^3}{(1 \text{ ft})^3} \right| \left| \frac{(0.0254 \text{ m})^3}{(1 \text{ in})^3} \right| \left| \frac{1 \text{ min}}{60 \text{ s}} \right| = \frac{1.19 \text{ m}^3}{\text{s}}$$

G. 75 mile/h to km/h

1 mile = 5280 ft $\therefore$ 12 in = 1 ft $\therefore$ 1 in = 0.0254 m $\therefore$ 1000 m = 1 km

$$\frac{75 \text{ mile}}{\text{h}} \left| \frac{5280 \text{ ft}}{1 \text{ mile}} \right| \left| \frac{12 \text{ in}}{1 \text{ ft}} \right| \left| \frac{0.0254 \text{ m}}{1 \text{ in}} \right| \left| \frac{1 \text{ km}}{1000 \text{ m}} \right| = \frac{120.7 \text{ km}}{\text{h}}$$

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$12 \cdot 8 + 1 \cdot 18 = 114.23 \text{ g/mol}$ $\leftarrow$ molar mass of octane

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$W = mg$ $\leftarrow$ we know force is equal to mass times gravitational force

$$W = (34.269 \text{ kg})(9.81 \text{ m/s}^2)$$

$$W = 336.19 \text{ N}$$

4 Question 4 10 / 10

✓ - **0 pts** Correct

- **5 pts** Redo calculation of circled parts.

- **5 pts** Did not answer 4 b)

- **10 pts** No answer

B. molar based specific volume

$$\frac{5 \text{ m}^3}{0.3 \text{ kmol}} = 16.66 \frac{\text{m}^3}{\text{kmol}}$$

mass based specific volume

$$\frac{5 \text{ m}^3}{34.269 \text{ kg}} = 0.146 \frac{\text{m}^3}{\text{kg}}$$

#5.

$$P = A + \left(\frac{B}{V} \right)$$

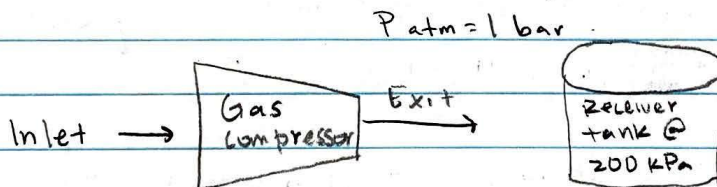
$$V = \text{ft}^3 \quad ; \quad P = \frac{\text{lb}_f}{\text{ft}^2}$$

$$\frac{\text{lb}_f}{\text{ft}^2} = A + \frac{B}{\text{ft}^3}$$

$$A = \frac{\text{lb}_f}{\text{ft}^2}$$

$$B = \text{lb}_f \cdot \text{ft}$$

#6.



$$1 \text{ kPa} = 0.01 \text{ bar}$$

$$\frac{200 \text{ kPa}}{1 \text{ kPa}} \bigg| \frac{0.01 \text{ bar}}{1 \text{ kPa}} = 2 \text{ bar}$$

$$p(\text{gage}) = p(\text{absolute}) - P_{\text{atm}}(\text{absolute})$$

$$p(\text{gage}) = 2 - 1 = 1 \text{ bar} = 100 \text{ kPa}$$

gage pressure because the pressure of the system is greater than the local atmospheric pressure.

5 Question 5 10 / 10

✓ - 0 pts Correct

- 5 pts Unit of B is wrong.
- 0 pts You lost a "f" on A
- 5 pts Unit of B is wrong. $[B]=[P]*[V]$
- 2.5 pts So what's the unit of B?
- 10 pts No answer submitted.

B. molar based specific volume

$$\frac{5 \text{ m}^3}{0.3 \text{ kmol}} = 16.66 \frac{\text{m}^3}{\text{kmol}}$$

mass based specific volume

$$\frac{5 \text{ m}^3}{34.269 \text{ kg}} = 0.146 \frac{\text{m}^3}{\text{kg}}$$

#5.

$$P = A + \left(\frac{B}{V} \right)$$

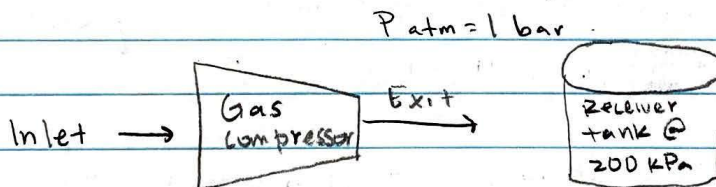
$$V = \text{ft}^3 \quad ; \quad P = \frac{\text{lb}_f}{\text{ft}^2}$$

$$\frac{\text{lb}_f}{\text{ft}^2} = A + \frac{B}{\text{ft}^3}$$

$$A = \frac{\text{lb}_f}{\text{ft}^2}$$

$$B = \text{lb}_f \cdot \text{ft}$$

#6.



$$1 \text{ kPa} = 0.01 \text{ bar}$$

$$\frac{200 \text{ kPa}}{1 \text{ kPa}} \bigg| \frac{0.01 \text{ bar}}{1 \text{ kPa}} = 2 \text{ bar}$$

$$p(\text{gage}) = p(\text{absolute}) - P_{\text{atm}}(\text{absolute})$$

$$p(\text{gage}) = 2 - 1 = 1 \text{ bar} = 100 \text{ kPa}$$

gage pressure because the pressure of the system is greater than the local atmospheric pressure.

6 Question 6 10 / 10

✓ - 0 pts Correct

- 5 pts Wrong judgement on vacuum and gage pressure.
- 5 pts Wrong reading of Bourdon gage.
- 10 pts No answer submitted.

#7

$$\Delta p = \rho g L$$

ρ : density of the manometer liquid
 g : acceleration of gravity
 L : difference in liquid levels

specific volume: $0.00122 \text{ m}^3/\text{kg}$

specific volume = $\frac{1}{\rho}$ 1.44 kPa

$$0.00122 \text{ m}^3/\text{kg} = \frac{1}{\rho} \quad \rho = 819.67 \text{ kg/m}^3$$

$$\Delta p = (819.67 \text{ kg/m}^3)(9.81 \text{ m/s}^2)(0.12 \text{ m})$$

$$\Delta p = 964.92 \text{ Pa} \div 1000 = 0.965 \text{ kPa}$$

The pressure decreases as the kerosene flows from point A to B as the pipe's diameter is decreased.

7 Question 7 10 / 10

- **2 pts** Wrong units of final results.

✓ - **0 pts** correct

- **3 pts** Wrong judgement of pressure increase or decrease.

- **3 pts** Did not answer the second question, increase or decrease of pressure.

- **7 pts** Did not answer the pressure difference between a and b.

- **10 pts** No answer.

Roxanna Mendoza

CHE 201

Homework 1

Question 8.

A sphygmomanometer is a device used to make measurements of one's diastolic and systolic blood pressure. First, the brachial artery is constricted (no blood flow) via an inflatable cuff. As blood begins to flow into the aorta after being constricted, a loud sound is heard through a stethoscope placed underneath the cuff.

Right when the sound is first heard the pressure shown by the gauge is recorded. This pressure is referred to as the systolic pressure (usually less than 120 mmHg). This sound is caused when turbulent blood flow hits the arterial walls. The sound is amplified when the motion of the arterial walls matches its resonant frequency.

As the cuff releases its grip, blood flow further increases. Blood flow is laminar and there is no sound heard through the stethoscope. This pressure is referred to as diastolic pressure (usually lower than 60 mmHg).

These values are significant as they allow a physician to know the current health condition of a patient. If both of these measurements are higher than usual, it is said that the patient has high blood pressure which can lead to a higher risk of strokes or heart attacks.

If the systolic measurement is high and the diastolic measurement remains normal, it is referred to as isolated systolic hypertension (ISH). ISH can indicate health conditions such as anemia, diabetes and hypertelorism.

If diastolic pressure is low and the systolic measurement remains normal, it is referred to as isolated diastolic hypotension (IDH). Patients with IDH are at a higher risk of coronary artery disease.

Works Cited

- Babbs, Charles F. "The Origin of Korotkoff Sounds and the Accuracy of Auscultatory Blood Pressure Measurements." *Journal of the American Society of Hypertension : JASH*, U.S. National Library of Medicine, Dec. 2015, www.ncbi.nlm.nih.gov/pubmed/26553392.
- Jr., William C. Shiel. "Definition of Sphygmomanometer." *MedicineNet*, MedicineNet, 11 Dec. 2018, www.medicinenet.com/script/main/art.asp?articlekey=15771.
- Seladi-Schulman, Jill. "High Systolic Blood Pressure: Causes and Treatments." *Healthline*, Healthline Media, 7 Dec. 2018, www.healthline.com/health/heart-disease/high-systolic-blood-pressure.
- "Sphygmomanometer." *Practical Clinical Skills*, www.practicalclinicalskills.com/sphygmomanometer.

8 Question 8 20 / 20

✓ - 0 pts Well explained.

- 4 pts Investigation on the physical basis of Korotkoff sound could be more detailed and specific.

- 4 pts Investigation on the role of systolic and diastolic pressure on medical practice could be more detailed and specific.

- 2 pts Investigation on the physical basis of Korotkoff sound could be more detailed and specific.

- 2 pts Investigation on the role of systolic and diastolic pressure on medical practice could be more detailed and specific.

- 2 pts No references listed.

- 20 pts No answer.

- 8 pts No answer of the significance of these pressures on medical practice.

- 10 pts The handwriting is hard to read. Also, the investigation could be more detailed and specific.